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The Hydro-Meteorological Data Analysis on the Impacts of Climate Variability on Water Resources Governance in the Lake Naivasha Basin, Kenya

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management.*

This study presents a comprehensive hydro-meteorological analysis of the Lake Naivasha Basin, a critical freshwater ecosystem in Kenya's Rift Valley, to evaluate the temporal and spatial variability of precipitation, temperature, evaporation, and lake water levels over multiple decades. The analysis integrates primary and secondary datasets obtained from the Kenya Meteorological Department, the Water Resources Authority, and satellite observations. Rigorous data quality control procedures were applied, including interpolation-based gap filling for missing precipitation and temperature records, outlier detection using visual inspection and Z-score statistics, and homogeneity assessment through the Standard Normal Homogeneity Test (SNHT) and Pettit's test. Spatial datasets were standardised to the WGS 84 UTM Zone 36S coordinate reference system to ensure consistency in spatial analyses. Time-series analyses were conducted using monthly and annual aggregations to examine seasonal and interannual hydro-climatic variability. The results reveal pronounced seasonal and long-term fluctuations in rainfall and temperature, indicating the influence of climate variability and ongoing climate change. A strong association was observed between precipitation variability and lake-level dynamics, with hydrological responses further intensified by anthropogenic pressures such as irrigation expansion and land-use change. These findings enhance understanding of the hydro-meteorological processes governing the Lake Naivasha Basin and provide valuable evidence to support sustainable water resources management, climate change adaptation, and informed regional planning.

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INTRODUCTION

The Lake Naivasha Basin is one of Kenya's important freshwater systems and is located within the eastern arm of the Great Rift Valley. The basin supports a wide range of ecological, economic, and social activities, including irrigated agriculture, horticultural production, tourism, domestic water supply, fisheries, biodiversity conservation, and other livelihood activities. Lake Naivasha and its associated rivers and wetlands therefore constitute an important water-resource system whose sustainability is closely linked to both human well-being and ecosystem functioning. The basin is also recognised as an important ecological system, with its wetlands providing habitat for diverse aquatic and terrestrial species while supporting economically significant activities around the lake. Consequently, changes in the availability and quality of water within the basin have implications that extend beyond hydrological processes to include food production, livelihoods, ecosystem conservation, and regional economic development.

The hydrological functioning of the Lake Naivasha Basin is strongly influenced by hydro-meteorological conditions, particularly precipitation, temperature, evaporation, river inflows, and changes in lake levels. These variables determine the quantity, timing, and spatial distribution of water entering, being stored within, and leaving the basin. Variability in rainfall and river discharge can consequently produce substantial changes in lake storage and water availability. A long-term assessment by Odongo et al. (2015), covering approximately five decades of observations, demonstrated considerable hydro-climatological variability

within the basin and identified a significant decline in observed lake volume. Importantly, the study found that observed changes in precipitation alone could not adequately explain the decline in lake volume and minimum river flows, suggesting that human-induced changes in land cover, infrastructure, groundwater interactions, and water abstraction may also contribute to changes in the basin's hydrological regime.

The importance of hydro-meteorological variability is further demonstrated by research examining extreme streamflow conditions in the basin. Kyambia and Mutua (2014) analysed streamflow records from 1959 to 2008 and reported significant changes in several indicators of extreme flows. Such changes have implications for water-resource planning because variations in the frequency and magnitude of high and low flows can affect water supply, agricultural production, ecosystem requirements, flood risk, and the reliability of water-resource infrastructure. These findings demonstrate that understanding both average conditions and extremes is essential for evaluating the changing characteristics of the basin's water resources.

In addition to climate and hydrological variability, the Lake Naivasha Basin has undergone substantial land-use and land-cover changes associated with agricultural expansion and intensification. These changes are particularly important because they can alter evapotranspiration, runoff generation, groundwater recharge, and the overall basin water balance. Odongo et al. (2019) demonstrated that transitions between grassland and cropland influenced evapotranspiration in the Lake Naivasha Basin, with newly cultivated areas

increasing local water demand. Their findings emphasise that changes in land use can modify hydrological processes and therefore need to be considered alongside climatic factors when assessing changes in water availability.

Water abstraction constitutes another important pressure on the basin's water resources. Lake Naivasha supports intensive agricultural and industrial activities, while groundwater is also utilised to meet increasing water demands. Yihdego et al. (2016) demonstrated the importance of groundwater-surface-water interactions in determining the basin's water balance and estimated substantial abstraction from both groundwater and lake water. Their analysis further showed that anthropogenic abstraction can have important consequences for lake levels and the overall water balance. Thus, changes in lake levels should be understood as the outcome of interactions among climatic inputs, surface-water inflows, evaporation, groundwater processes, abstraction, land-use change, and other human interventions rather than as a direct response to rainfall variability alone.

These interacting pressures create significant challenges for sustainable water-resources management in the basin. Water demand from agriculture, domestic users, industry, tourism, and ecological systems must be accommodated within a freshwater system characterised by substantial temporal variability in water availability. During periods of low rainfall and reduced river inflows, competition among water users may intensify, increasing pressure on water-allocation systems and potentially creating conflicts among users. Conversely, periods of exceptionally high rainfall and river discharge may increase the risks of flooding, erosion, sediment transport, and damage to water infrastructure. Effective management therefore requires an understanding of both water availability and the factors that control its temporal and spatial variability.

The relationship between hydro-meteorological variability and water governance is particularly important in the Lake Naivasha Basin. Water governance encompasses the institutions, policies,

regulations, allocation mechanisms, monitoring systems, and stakeholder processes through which water resources are managed and shared. Ndungu et al. (2014) identified knowledge gaps and institutional challenges associated with integrated water resources management (IWRM) in the Lake Naivasha Basin and emphasised the importance of integrating scientific information with stakeholder knowledge to support sustainable governance. This suggests that effective governance cannot be separated from reliable information concerning the basin's hydrological and climatic conditions.

A key challenge is that water-resource management systems may be influenced by assumptions based on historical patterns of water availability. However, increasing hydro-meteorological variability and uncertainty can make such assumptions less reliable. Changes in rainfall seasonality, temperature, evaporation, streamflow, and lake levels may alter both water supply and demand. Higher temperatures, for example, can increase evaporative losses and water requirements for agriculture, while changes in rainfall intensity and seasonality can modify runoff and river inflows. Consequently, water-resource planning based exclusively on historical averages may not adequately account for emerging variability and changing hydrological conditions.

The available evidence indicates that the Lake Naivasha Basin requires an integrated assessment of climatic, hydrological, land-use, and anthropogenic processes. Odongo et al. (2015) found that precipitation changes alone could not account for the observed long-term decline in lake volume, while Odongo et al. (2019) demonstrated the influence of land-use transitions on evapotranspiration. Similarly, Yihdego et al. (2016) highlighted the importance of groundwater and water abstraction in the basin's water balance. Collectively, these studies demonstrate that the basin's water resources are controlled by a complex interaction of natural and human factors. However, there remains a need for integrated analysis that explicitly connects observed hydro-meteorological changes with their

implications for water-resource governance and environmental planning.

Against this background, the present study examines historical and recent hydro-meteorological trends and variability in the Lake Naivasha Basin and assesses their implications for water-resource governance and environmental planning. The study focuses on precipitation, temperature, evaporation, river inflows, and lake-level fluctuations to establish the temporal and spatial characteristics of the basin's hydro-meteorological system. It further investigates the relationships among these variables and considers how their variability influences water availability and the sustainability of water resources.

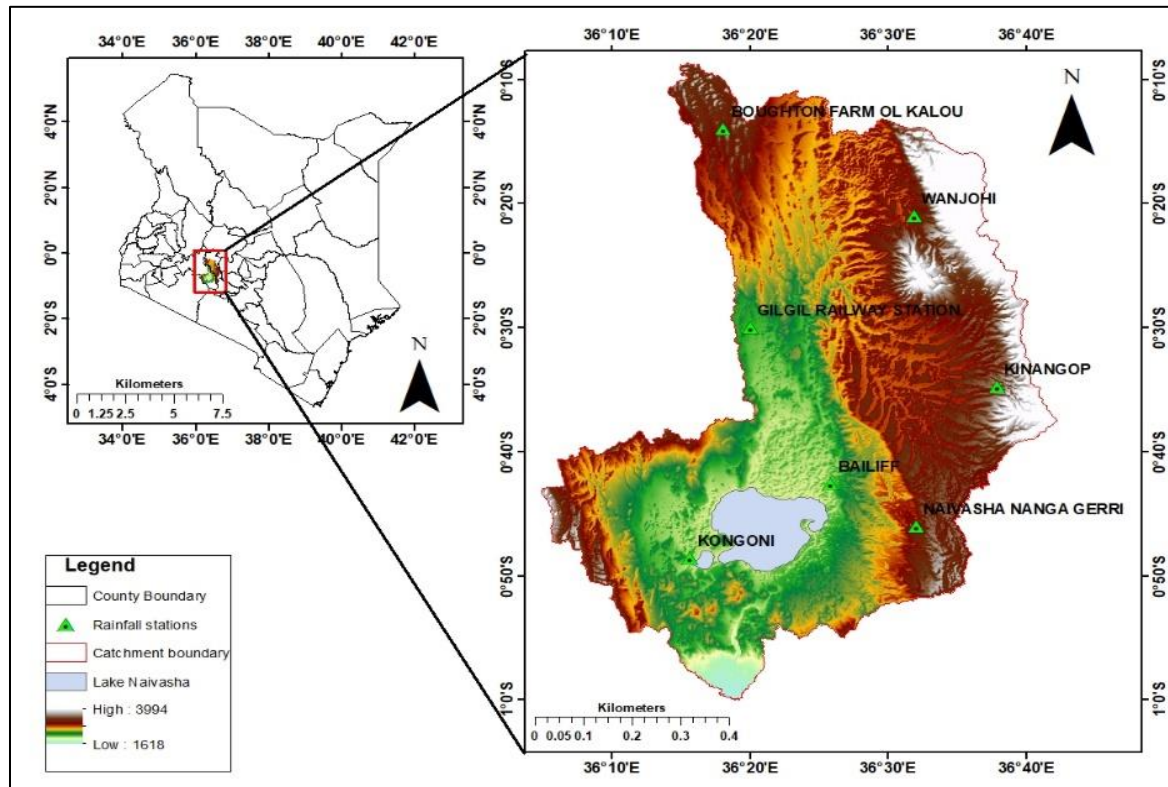
The study is guided by the recognition that hydro-meteorological processes and water governance are closely interconnected. Changes in the quantity and timing of water availability can influence water allocation, competing water uses, ecosystem requirements, monitoring, conservation, and environmental planning. Conversely, human activities and management interventions can modify hydrological processes and affect the water balance of the basin. The study therefore seeks to move beyond a description of climatic and hydrological trends by examining their implications for the management and governance of a water system subject to increasing environmental and socio-economic pressures.

METHODS

The Study Area

Lake Naivasha Basin is located in the central part of the Kenyan Rift Valley, approximately 100 kilometres northwest of Nairobi. The basin lies between latitudes $0^{\circ}45'S$ and $1^{\circ}15'S$ and longitudes $36^{\circ}00'E$ and $36^{\circ}45'E$, covering an estimated area of about 3,400 km². Lake Naivasha itself is a freshwater lake situated at an altitude of approximately 1,884 meters above sea level, making it one of the few freshwater bodies in the eastern Rift Valley. The physical and Hydrometeorological features in the study area include the basin, which is characterised by a complex hydrological system that includes several perennial and seasonal rivers, the most significant being the Malewa, Gilgil, and Karati Rivers, all of which contribute over 90% of the lake's surface inflow. The Lake is endorheic, with no visible surface outlet, and water loss occurs primarily through evaporation and groundwater seepage, which plays a significant role in the region's hydrogeology. One hydroclimatological assessment estimated the lake area at approximately **169 km²** and the basin area at about **3,500 km²**, while more recent satellite-based assessments report a mean lake area of approximately **129 km² for 1984–2020**, demonstrating the substantial temporal variability in the lake's extent (Odongo et al., 2015; Midekisa et al., 2021).

Figure 1: Map of Lake Naivasha Basin



Climate

The basin lies in the Kenyan Rift Valley, with altitudes ranging from about 1 890 m (at the lake surface) up to ~3 990 m at some ridges of the catchment, with a total basin area of about ~3 500 km², with the lake covering around ~169 km². Rainfall within the basin is normally influenced by relief, such as the wetter slopes of Aberdare Mountains versus drier lowlands, and the basin lies partially in a rain- shadow. The climate in the Lake Naivasha Basin is classified as semi-arid to sub-humid, with bimodal rainfall patterns, typically occurring from March to May (long rains) and October to December (short rains). Annual rainfall in the basin ranges significantly: lowland areas receive about ~610 mm per year, while wetter slopes get up to ~1 525 mm per year. This varies significantly across the basin, ranging from about 600 mm in the lowlands to over 1,200 mm in the highland catchments. It is significant to note that rainfall is bimodal: a “long-rains” season roughly March–June (MAMJ) and a “short-rains” season roughly October–December (OND). The climate classification is therefore semi-arid to dry in parts: The lake site itself experiences an average

maximum temperature of ~26.5 °C, a mean minimum of ~16.4 °C; average rainfall of ~600 mm, and the Mean annual temperature ranges from 15°C to 25°C, with higher elevations experiencing cooler conditions. Potential evaporation (or evapotranspiration) in the basin is very high relative to rainfall: values of ~1 400 mm and even up to ~1 525 mm in dry months, making the water balance quite “tight”: the rainfall covers not only runoff but large evaporation losses; this causes a serious challenge of high evaporation and relief-induced heterogeneity and spatial and temporal variability in water input. Additionally, high evaporation and relatively modest rainfall in lowland zones mean the lake and its inflows are sensitive to both rainfall variability and catchment changes, including land-use and abstraction.

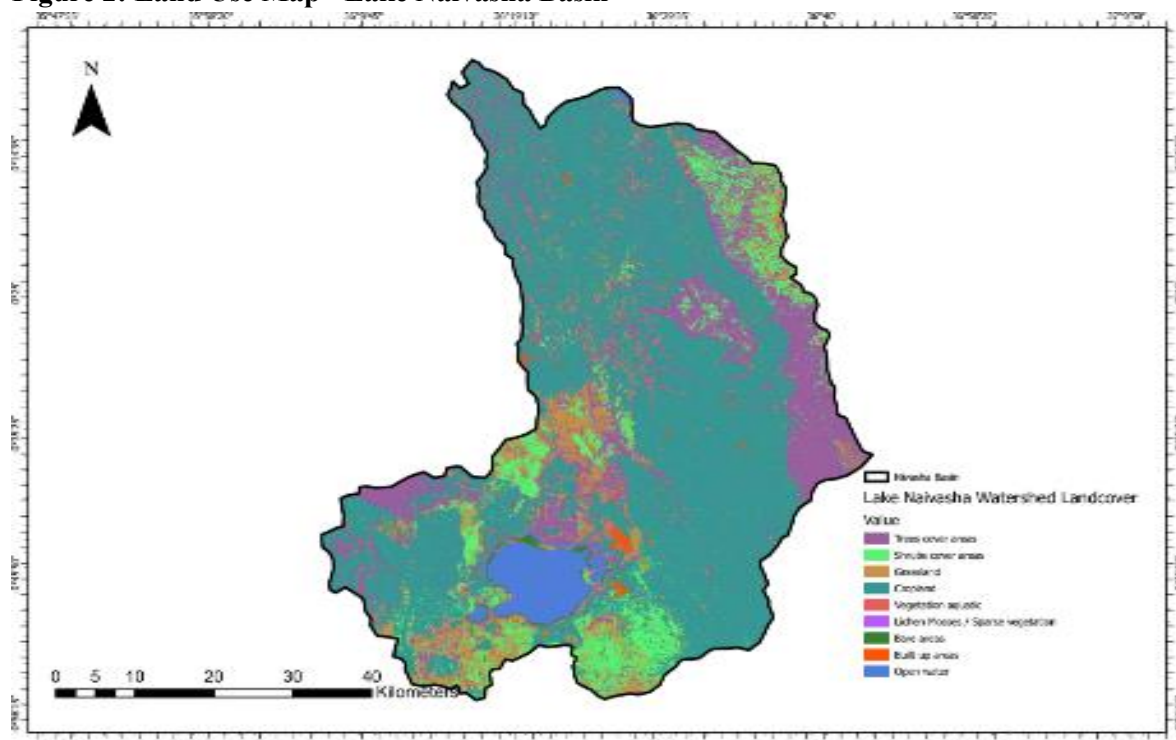
Land Use and Economic Activities

The basin supports a wide range of land uses, including: Agriculture (both rain-fed and irrigated, especially flower farming near the lake); Livestock grazing; Forestry in the upper catchments; and Urban development, particularly in Naivasha town. The lake and its surrounding wetlands also support a diverse range of flora and

fauna, making it a recognised Ramsar site and a hotspot for biodiversity and birdlife. However, land use changes, deforestation in the upper

catchments, and increasing water abstraction for agriculture have intensified pressure on the lake's hydrological balance.

Figure 2: Land Use Map - Lake Naivasha Basin



Socio-Economic Importance

Lake Naivasha is a critical resource for local livelihoods, providing water for domestic, agricultural, industrial, and ecological needs. The horticulture and floriculture industries, which rely heavily on irrigation from the lake and its tributaries, are among the most significant economic activities in the area, contributing substantially to Kenya's export earnings. Additionally, fishing and tourism are key livelihood sources for local communities.

DATA COLLECTION AND PROCESSING

The study employed a mixed-methods approach combining long-term hydro-meteorological and hydrological datasets with primary field investigations and qualitative information on water resources governance. To avoid ambiguity in the temporal scope of the study, the datasets were organised into distinct analytical periods according to their availability and purpose. The long-term hydro-meteorological and hydrological analysis covered the period 1957–

2022, providing a sufficiently long record for assessing trends, variability, and changes in the basin's water balance. The 2002–2010 period was used only where datasets with consistent and comparable observations were available and where the analysis required a common period across variables or stations; it was not treated as a separate overall study period. The 2017–2019 period refers specifically to primary field investigations and stakeholder consultations undertaken to obtain information on water use, institutional arrangements, governance practices, and stakeholder perceptions. Where satellite-derived land use/land cover data were used, the analysis covered selected years within the period for which suitable imagery was available rather than implying continuous annual observations.

Approximately ten field visits and stakeholder consultation sessions were undertaken between 2017 and 2019 across the Lake Naivasha Basin. The field component was designed to complement the long-term quantitative analysis by providing

contextual information on water abstraction, resource use, institutional arrangements, environmental pressures, and stakeholder experiences. Primary and secondary quantitative and qualitative data were therefore integrated to examine not only changes in the basin's hydro-meteorological system but also how these changes interact with water resources management and governance.

Secondary Hydro-Meteorological and Hydrological Data

Secondary datasets were obtained from authoritative institutions and established climate and remote-sensing databases. The principal hydro-meteorological variables comprised precipitation, minimum and maximum air temperature, evaporation, reference evapotranspiration, river discharge, and lake level. Additional spatial datasets, including land use/land cover, soil characteristics, topography, and drainage networks, were incorporated to support interpretation of catchment processes and hydrological responses.

Precipitation records were obtained from fourteen meteorological stations located within and around the Lake Naivasha Basin from the Kenya Meteorological Department (KMD). The available station records were used to characterise spatial and temporal rainfall variability and to identify long-term trends. Where station records contained gaps or did not provide sufficient spatial coverage, satellite-derived precipitation products, particularly the Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) and Tropical Rainfall Measuring Mission (TRMM) datasets, were used as supplementary sources. These products were not treated as direct replacements for station observations without evaluation; rather, their suitability was assessed by comparison with available ground observations before they were used to supplement incomplete records.

Daily minimum and maximum air temperature records were obtained from KMD meteorological stations. ERA5 reanalysis data were used to supplement spatial and temporal gaps in the

station records and to provide spatially consistent temperature information where observations were incomplete or unavailable. The temperature datasets were subsequently aggregated to monthly and annual scales for trend and variability analysis.

Evaporation observations were obtained from available hydrological monitoring stations. Where direct evaporation observations were unavailable or incomplete, atmospheric evaporative demand was represented through reference evapotranspiration (ET_o) estimated using the FAO Penman–Monteith method. The distinction between observed evaporation and estimated reference evapotranspiration was maintained throughout the analysis because the two variables represent different components of the hydrological system. ET_o was used primarily to characterise atmospheric evaporative demand, whereas lake and surface-water evaporation were considered in the water-balance assessment using the most appropriate available observations or estimates.

Daily and monthly river-discharge records for the principal surface-water inflows to Lake Naivasha—the Malewa, Gilgil, and Karati rivers—were obtained from the Water Resources Authority (WRA). These rivers were selected because they represent the principal monitored surface-water inflows to the lake. The discharge records were used to quantify seasonal and interannual variability in inflows and to investigate the relationship between rainfall variability, catchment response, and changes in lake level.

Historical Lake Naivasha water-level records were obtained from the Kenya Marine and Fisheries Research Institute (KMFRI) and the WRA. The lake-level records were used as an indicator of changes in lake storage and were analysed in relation to precipitation, river inflows, evaporation, and water abstraction. Where necessary, lake-level observations were converted into changes in storage using an appropriate lake area–elevation relationship rather than assuming

that a change in water level represents a directly proportional change in lake volume.

Land use and land cover (LULC) information was derived from Landsat and MODIS satellite imagery for selected representative years. The selection of years was based on image availability, cloud-free coverage, and the need to represent major stages of land-use change within the catchment. LULC analysis was used to identify changes in agricultural land, built-up areas, vegetation, wetlands, water bodies, and other relevant land-cover classes. These changes were subsequently considered when interpreting variations in runoff, water demand, and pressures on water resources.

Primary Data and Governance Information

Primary information was collected during field investigations and stakeholder consultations conducted between 2017 and 2019. The field component did not constitute a replacement for the long-term hydro-meteorological record; rather, it provided contemporary contextual information for interpreting the quantitative findings and assessing their implications for water resources governance.

Information on water-resource utilisation was obtained from the WRA and other relevant institutions. The information included locations of water abstraction points, water permits, authorised abstraction volumes, reported abstraction levels where available, sectoral water demands, and the principal categories of water use. These data were used to characterise the spatial and sectoral distribution of water demand and to identify areas where changes in water availability could potentially create or intensify competition among users.

Qualitative information on water governance was collected through stakeholder consultations, field observations, institutional records, and relevant policy and planning documents. Stakeholders were selected purposively based on their institutional responsibilities, resource use, or involvement in water management within the basin. The consultations focused on institutional

roles and responsibilities, water-allocation procedures, monitoring and enforcement, stakeholder participation, conflict and competition over water, environmental protection, and perceived changes in water availability. Particular attention was given to how stakeholders perceived the effects of rainfall variability, changing river flows, lake-level fluctuations, droughts, and other hydro-meteorological changes on water management decisions.

The governance component therefore examined both formal governance arrangements, including policies, institutions, regulations, permits, and allocation procedures, and practical governance responses, including stakeholder participation, monitoring, enforcement, conflict management, and adaptation to changing water availability.

Data Preprocessing and Quality Control

Prior to analysis, all hydro-meteorological, hydrological, and geospatial datasets underwent systematic quality-control and preprocessing procedures. These included screening for missing observations, duplicate records, transcription errors, inconsistencies, extreme values, and changes in measurement practices. The procedures were applied consistently across datasets to ensure that observed trends reflected environmental changes rather than errors or artificial discontinuities in the records.

Short-duration gaps in precipitation and temperature records were filled using interpolation only where the length of the missing period and the characteristics of the surrounding observations justified such treatment. Linear interpolation was used for short, isolated gaps, while longer gaps were supplemented using observations from neighbouring stations or validated gridded datasets. Interpolation was not applied indiscriminately to long periods of missing data because excessive infilling can artificially reduce the observed variability of a time series.

The quality of the datasets was evaluated using time-series plots, station-to-station comparisons, scatterplots, and statistical screening. Suspected

outliers were investigated against neighbouring observations and original station records before being retained, corrected, or excluded. Outliers were therefore not automatically removed because extreme precipitation, discharge, or lake-level observations may represent genuine hydrological events rather than measurement errors.

Temporal homogeneity was assessed using the Standard Normal Homogeneity Test (SNHT) and Pettitt's change-point test. These tests were used to identify possible discontinuities associated with station relocation, changes in instrumentation, modifications in observation practices, or other non-climatic influences. Where statistically significant breaks were identified, the affected records were examined in relation to station metadata and neighbouring observations before deciding whether adjustment or exclusion was appropriate.

The geospatial datasets were standardised using the WGS 84 / UTM Zone 36S coordinate reference system. Raster datasets were resampled to a common spatial resolution appropriate to the analysis, while vector datasets were checked for geometric and topological inconsistencies. Daily observations were aggregated into monthly and annual series for analyses requiring longer-term temporal scales. Seasonal aggregation was also undertaken to distinguish the main rainfall seasons and corresponding hydrological responses.

Hydro-Meteorological and Hydrological Analysis

The hydro-meteorological analysis focused on identifying the magnitude, direction, and significance of changes in the principal climatic and hydrological variables. Descriptive statistics, including means, standard deviations, coefficients of variation, minimum and maximum values, and seasonal distributions, were used to characterise temporal and spatial variability.

Long-term trends in precipitation, temperature, evaporation, river discharge, and lake levels were assessed using appropriate parametric and non-parametric statistical techniques. Where the assumptions of parametric methods were not

satisfied, non-parametric approaches were preferred because hydro-climatic datasets frequently exhibit skewness, serial correlation, and extreme observations. Trend magnitude was estimated using the Sen's slope estimator, while statistical significance was evaluated using the Mann–Kendall test, with appropriate consideration of serial dependence where necessary. Change-point analysis was used to identify periods during which substantial shifts in hydro-meteorological behaviour may have occurred.

Relationships among precipitation, temperature, evaporation, river discharge, and lake level were evaluated using correlation and regression analyses. These analyses were used to determine whether observed changes in lake levels and river inflows could be statistically associated with changes in climatic variables. The interpretation of these relationships considered the influence of water abstraction, land-use change, and other anthropogenic factors so that statistical associations were not automatically interpreted as direct causal relationships.

Water Balance and Groundwater Estimation

The Lake Naivasha water balance was formulated to quantify the principal components controlling changes in lake storage. The general water-balance equation was expressed as:

$$\Delta S = P + Q_{in} + G_{Win} - E - Q_{out} - G_{Wout} - W - \varepsilon$$

Where ΔS represents the change in lake storage; P is direct precipitation on the lake surface; Q_{in} represents surface-water inflows from the principal rivers; G_{Win} represents groundwater inflow; E represents evaporation from the lake surface; Q_{out} represents surface-water outflow; G_{Wout} represents groundwater outflow; W represents anthropogenic water abstraction; and ε represents the residual or error term.

Because Lake Naivasha has no major permanent surface outlet, the surface outflow term was treated according to the hydrological conditions of the lake and the availability of evidence for

episodic outflow. Lake storage changes were estimated from lake-level observations and an appropriate stage–area or stage–volume relationship. Direct precipitation on the lake was estimated from spatially averaged rainfall over the lake surface, while river inflows were derived from the monitored discharge records for the Malewa, Gilgil, and Karati rivers. Evaporation was estimated from available observations and/or appropriately derived evaporation estimates.

Groundwater was treated explicitly rather than being assumed to be negligible. Where direct groundwater-flow measurements were unavailable, groundwater contribution was estimated as a residual component of the water balance, subject to the limitations imposed by uncertainties in the other measured components. The residual was calculated only after accounting for precipitation, surface-water inflows, evaporation, lake-storage changes, and documented abstractions. Because residual groundwater estimates incorporate measurement and estimation errors from all other components, they were interpreted as an estimate of the net groundwater contribution and water-balance residual, rather than as a direct measurement of groundwater recharge or groundwater discharge. This distinction is important because a residual water balance cannot independently distinguish groundwater inflow from groundwater outflow unless additional groundwater observations are available.

Where groundwater data from boreholes, springs, piezometers, or groundwater-level monitoring points were available, these observations were used to assess the plausibility of the residual estimates. The groundwater analysis therefore combined available direct evidence with water-balance residual estimation rather than presenting the residual as an independently measured groundwater flux.

Linking Hydro-Meteorological Findings to Water Resources Governance

A central component of the methodology was the integration of the quantitative hydro-meteorological analysis with the qualitative

assessment of water resources governance. The linkage was established through a sequential analytical framework. First, the study identified statistically significant changes and periods of increased variability in precipitation, temperature, evaporation, river inflows, and lake levels. Second, these changes were translated into their implications for water availability, including changes in inflow reliability, lake storage, seasonal water availability, and evaporative losses. Third, the implications for water demand and allocation were examined using information on abstraction points, water permits, abstraction volumes, sectoral demands, and ecological requirements. Finally, the observed hydro-meteorological changes and water-availability conditions were compared with institutional arrangements, allocation procedures, monitoring systems, stakeholder experiences, and existing water-management policies.

This approach allowed the study to establish a clear pathway from hydro-meteorological change → water-resource response → water-use pressure → governance challenge → potential adaptive response. For example, evidence of declining or increasingly variable river inflows was examined in relation to the reliability of water available for abstraction and environmental requirements. Similarly, persistent lake-level fluctuations were considered in relation to abstraction pressure, ecosystem protection, and the capacity of institutions to adjust allocation and management decisions. Changes in rainfall and temperature were also examined in relation to drought risk, agricultural water demand, and the need for improved monitoring and adaptive planning.

The qualitative stakeholder data were analysed thematically around key governance dimensions, including water allocation, institutional coordination, monitoring and enforcement, stakeholder participation, conflict and competition, ecosystem protection, climate adaptation, and implementation of water policies. Themes emerging from stakeholder consultations were then compared with the quantitative hydro-meteorological findings. This triangulation enabled the study to determine whether

stakeholder-reported changes in water availability and management challenges were consistent with observed hydro-meteorological trends and hydrological responses.

The integration of these datasets was intended to avoid treating governance as a separate descriptive component of the study. Instead, governance was examined as the institutional mechanism through which hydro-meteorological changes are translated into decisions about water allocation, use, conservation, monitoring, and adaptation. The findings were consequently used to assess whether existing governance arrangements are sufficiently flexible to respond to observed and emerging changes in the basin's hydro-

meteorological conditions and to identify areas where adaptive and integrated water resources management may be strengthened.

Hydro-Meteorological Data Components

The hydro-meteorological database comprised observations describing the interactions between atmospheric processes and the basin's surface and subsurface water systems. These datasets represented the principal meteorological and hydrological variables controlling the Lake Naivasha water balance and were obtained from field measurements, institutional databases, remote sensing products, and derived hydrological computations.

Table 1: Hydro-meteorological Datasets Used in the Study.

Category	Parameters	Measurement Instruments/Data Sources
Meteorological	Rainfall, minimum and maximum air temperature, relative humidity, wind speed and direction, solar radiation	Rain gauges, automatic weather stations, thermometers, hygrometers, anemometers, pyranometers, satellite observations (CHIRPS, TRMM, ERA5)
Hydrological	River discharge, stream stage, groundwater level, lake water level, evaporation	Stream gauging stations, staff gauges, pressure transducers, piezometers, evaporation pans, lysimeters, remote sensing observations
Derived Variables	Surface runoff, infiltration, evapotranspiration, water balance components	Estimated using hydrological models, GIS-based analyses, and established empirical and physically based equations (e.g., FAO Penman–Monteith method)

The collection, processing, interpretation, and analysis of these datasets involved the integration of statistical, geospatial, and time-series analytical techniques. These complementary approaches enabled the assessment of long-term trends, seasonal variability, change points, and interrelationships among key hydro-meteorological variables, including precipitation, temperature, river discharge, evapotranspiration, and lake water levels. The integrated dataset provided the foundation for evaluating the impacts of climate variability and anthropogenic pressures on hydrological processes and water resources within the Lake Naivasha Basin. To evaluate the temporal consistency and homogeneity of the

climate time series, the Standard Normal Homogeneity Test (SNHT) and Pettitt’s test were applied. These tests helped detect potential shifts or discontinuities in the data resulting from changes in measurement techniques, station relocations, or environmental factors. All spatial datasets were projected into a uniform coordinate reference system, WGS 84 / UTM Zone 36S, to ensure spatial comparability across the study area. Finally, temporal aggregation of time series data was performed at monthly and annual scales to facilitate the assessment of seasonal and interannual variability in climatic and hydrological parameters.

Table 2: Temporal Aggregation of Time Series Data

Category	Parameters	Instruments/Sources
Meteorological	Rainfall, Temperature, Humidity, Wind speed/direction, Solar radiation	Rain gauges, Thermometers, Hygrometers, Anemometers, Satellite data
Hydrological	Streamflow, River stage, Groundwater /Lake level, Evaporation.	Stream gauges, Piezometers, Lysimeters, Remote sensing
Derived data	Runoff, Infiltration, Water balance, Evapotranspiration	Computed using hydrological models and equations

The collection, processing, interpretation, and application of data related to the interaction between the atmosphere and the Earth's surface water systems, including precipitation, temperature, humidity, wind, evaporation, and stream for the Lake Naivasha Basin, involved a combination of statistical, geospatial, and time series analysis techniques.

These methods were selected to assess trends, variability, and relationships between key hydro-meteorological variables, specifically rainfall, temperature, streamflow, evapotranspiration, and lake levels. The key components of hydro-meteorological data from multiple sources that represented both meteorological and hydrological variables included:

Merging of Station Observations with CHIRPS/TRMM/ERA5 Data

Key Components of Hydro-Meteorological Data

Hydrometeorological observations from the Kenya Meteorological Department (KMD) and Water Resources Authority (WRA) were merged with CHIRPS, TRMM, and ERA5 datasets to improve spatial and temporal climate coverage in the Lake Naivasha Basin. Station data underwent quality control to remove inconsistencies, outliers, and duplicate records, while minor missing values were filled through interpolation using neighbouring stations. Station coordinates were matched with corresponding gridded datasets, and all datasets were harmonised to a common monthly temporal resolution. Consistency between station and gridded data was evaluated using statistical indicators including correlation coefficient (R), Root Mean Square Error (RMSE), Mean Bias Error (MBE), and Nash–Sutcliffe

Efficiency (NSE). Where biases were identified, correction factors based on station observations were applied to the gridded datasets before merging through weighted averaging/interpolation. The final merged dataset was validated using independent station records, showing improved agreement between observed and modelled hydroclimatic patterns across the basin.

Trend Analysis

Data Availability and Representation of Trend Analysis

Hydrometeorological data used in this study were obtained from the Kenya Meteorological Department (KMD) and the Water Resources Authority (WRA) for the period 2002–2010. The dataset comprised rainfall, temperature, streamflow, and evapotranspiration records collected from stations distributed across the Lake Naivasha Basin. Data availability was assessed based on the number of operational stations, record length, and percentage of missing observations. Rainfall records had the widest spatial coverage, with most stations providing continuous monthly observations over the 2002–2010 analysis period and less than 10% missing data after quality control. Temperature and evapotranspiration datasets showed moderate completeness, with occasional gaps mainly associated with instrument downtime and reporting inconsistencies. Streamflow records from gauging stations exhibited comparatively higher missing values in some years due to interrupted monitoring activities. Only stations with sufficiently long and reliable records were retained for trend analysis to ensure

representativeness and consistency. Missing observations were minimised through quality control and interpolation using neighbouring stations where gaps were small and non-systematic. The selected 2002–2010 period provided adequate temporal coverage for detecting hydroclimatic variability and assessing Trends. The Long-term trends in rainfall, temperature, and streamflow were analysed using the non-parametric Mann-Kendall trend test and Sen's slope estimator.

Mann–Kendall Trend Analysis Implementation

Trend analysis of hydroclimatic variables was conducted using the Mann–Kendall (MK) non-parametric test at a significance level of $\alpha=0.05$ (95% confidence interval). The test was selected because it is robust for detecting monotonic trends in hydrometeorological time series that may not satisfy normality assumptions. Prior to applying the MK test, the datasets were examined for serial autocorrelation using lag-1 autocorrelation analysis. Where significant autocorrelation was detected, a modified Mann–Kendall approach with pre-whitening was applied to minimise the influence of serial dependence on trend significance. This adjustment reduced the likelihood of overestimating statistically significant trends due to persistence within the time series. The magnitude of detected trends was estimated using Sen's slope estimator, which provided the rate of change per unit time for rainfall, temperature, and streamflow variables over the 2002–2010 study period. Statistical analyses were performed using standard hydrological and climatological trend analysis procedures to ensure consistency and reliability of results.

Rainfall-runoff relationships were analysed using runoff coefficients and simple water balance approaches to assess how rainfall variability affects river discharge and lake levels. Climate variability assessment: Temporal distribution of extreme events (e.g., droughts, floods) was examined using standardised indices such as the Standardised Precipitation Index (SPI) and

Standardised Streamflow Index (SSI). Lake Naivasha lies in the East African Rift, and its basin is subject to significant hydrological, climatic, and land-use pressures, such as agriculture, floriculture, abstraction, wetlands loss, that affect water quantity and quality, rainfall, streamflow (inflows/outflows), lake level or volume, and minimum flows, which are key for understanding how the system is changing for water resources management. Trend analysis helps show if rainfall is increasing or decreasing, changing flows, and whether the lake levels are rising or falling. To detect long-term changes in climatic and hydrological variables, various statistical analysis were applied, including the Mann-Kendall Trend Test; a non-parametric test used to detect monotonic trends in time series data without assuming a specific distribution and the Sen's Slope Estimator the was employed alongside the Mann-Kendall test to quantify the rate of change over time and to establish the annual and seasonal rainfall trends, Minimum and maximum temperatures and the trends in river discharges and lake levels.

Correlation and Regression Analysis

This was undertaken using the Pearson and Spearman correlation coefficients in assessing the relationships between Rainfall and river inflows, river discharge, lake levels, temperature, and evapotranspiration. Simple linear regression was used to determine the strength and direction of relationships between dependent and independent variables. The Pearson and Spearman correlation coefficients were calculated to examine relationships among rainfall, river inflows, lake levels, and evapotranspiration.

Water Balance Estimation

A basic water balance approach was used to evaluate the lake's hydrological response. This helped understand the interaction between inflow, rainfall, and lake level variability:

$$\Delta S = P + Q_{in} - ET - Q_{out} - G \quad \Delta S = P + Q_{in} - ET - Q_{out} - G$$

Where:

- ΔS : Change in lake storage
- PPP: Precipitation on the lake surface
- Q_{in} : Surface inflows from rivers
- ET: Evapotranspiration losses
- Q_{out} : Surface outflows (negligible due to no visible outlet)
- GGG: Groundwater exchange (estimated qualitatively).

Limitations in Water Balance and Groundwater–Lake Interactions

Although the basin water balance incorporated major hydrological components including precipitation, evapotranspiration, surface inflows, and streamflow, constraints remain regarding the quantification of groundwater–lake exchanges due to limited long-term groundwater monitoring data within the Lake Naivasha Basin. Consequently, groundwater inflow and outflow processes were inferred indirectly from observed lake level variations, hydrological responses, and findings from previous basin-scale hydrogeological studies. The analysis therefore constrains the plausible direction and relative magnitude of groundwater–lake interactions rather than providing direct volumetric estimates.

Existing studies within the basin indicate that Lake Naivasha is hydraulically connected to surrounding aquifer systems, with groundwater contributing to lake recharge during wetter periods and potential seepage losses occurring during dry conditions. However, uncertainties associated with subsurface heterogeneity, abstraction rates, and limited piezometric observations restrict precise quantification of these exchanges. Accordingly, interpretations of groundwater contributions in this study are presented as indicative basin-scale estimates intended to support understanding of hydrological variability and water resource governance, rather than as definitive groundwater flux measurements.

Climate Variability and Extremes

To assess climatic variability and extremes within the study area, a suite of standardised indices and

statistical methods was employed. The Standardised Precipitation Index (SPI) was utilised to characterise meteorological droughts at multiple time scales (1-month, 3-month, and 12-month). The SPI enabled the quantification of precipitation anomalies relative to long-term averages, thereby identifying short-term and long-term dry and wet conditions. For the assessment of hydrological droughts, the Standardised Streamflow Index (SSI) was applied to river discharge data. This index facilitated the detection of prolonged low-flow periods and their correspondence with meteorological drought events.

In addition, frequency analysis was conducted to evaluate extreme rainfall and flood events. Rainfall percentiles were used to define thresholds for extreme events, while return period estimation was performed using the Gumbel Extreme Value Distribution to quantify the likelihood and recurrence of high-intensity rainfall and flood occurrences. These combined approaches provided a comprehensive understanding of the spatial and temporal patterns of climate variability, drought severity, and extreme hydro-meteorological events within the study basin.

Spatial Analysis Using GIS and Remote Sensing

The application of GIS and remote sensing tools provided critical insights into the spatial dynamics of the Lake Naivasha Basin. ArcGIS was employed to map and visualise the spatial variation in rainfall, topography, and land use, highlighting areas most susceptible to hydrological stress. The use of Google Earth Engine enabled efficient analysis of vegetation dynamics and land cover changes over time, particularly through the computation of the Normalised Difference Vegetation Index (NDVI). This facilitated the assessment of temporal trends in vegetation health and density, offering a proxy for ecosystem condition and human impacts.

Land use and land cover analyses revealed significant anthropogenic influences on the basin's hydrology, including deforestation and agricultural expansion. These changes were

spatially correlated with alterations in rainfall distribution and surface water extent, suggesting that human activities are a key driver of local hydrological variability. Mapping surface water dynamics through remote sensing further illustrated seasonal and interannual fluctuations in lake and wetland extents, which are critical for water resource planning and ecosystem management. Overall, the integration of GIS and remote sensing techniques provided a powerful framework for understanding the spatial heterogeneity of hydrological and ecological processes in the Lake Naivasha Basin. These tools not only quantify human-induced changes but also support the development of targeted management strategies, enabling more informed decision-making for sustainable water resource and land management in the region.

RESULTS AND DISCUSSIONS

The principal period adopted for the long-term hydro-meteorological and hydrological trend analysis was **1957–2022**, providing an extended record for examining temporal variability, trends, and changes in the hydro-climatic characteristics of the Lake Naivasha Basin. The use of a long observation period is particularly important in hydro-climatic studies because it allows short-term interannual variability to be distinguished from persistent long-term trends and potential shifts in the hydrological regime. Previous research on the Lake Naivasha Basin has similarly demonstrated the value of long-term records, with Odongo et al. (2015) analysing approximately five decades of precipitation, river-flow, and lake-volume data for the period 1960–2010 to investigate hydro-climatological trends, variability, and changes in the basin water balance. Their findings demonstrated substantial variability in hydro-climatic conditions and a significant long-term decline in lake volume, while also indicating that climatic variability alone could not explain all observed hydrological changes.

Because the individual datasets used in this study differed in their periods of record, availability, and continuity, specific sub-periods were analysed

where appropriate. In particular, the period **2002–2010** was used only for analyses requiring a common temporal window among variables or datasets with shorter overlapping records. This shorter period therefore represents a **dataset-specific analytical window** rather than a replacement for the principal 1957–2022 study period. Such differentiation is important because hydro-meteorological variables and hydrological observations frequently have different lengths and levels of temporal continuity, requiring common-period analyses when relationships among variables are examined.

The period **2017–2019** was reserved for primary field investigations, stakeholder consultations, and collection of contemporary information concerning water use, governance arrangements, institutional practices, and stakeholder perceptions. These field-based data complement the long-term hydro-meteorological analysis by providing information on contemporary water-resource management conditions and stakeholder experiences. Accordingly, the different periods reported in the study should be understood as serving **distinct analytical purposes**: 1957–2022 represents the principal long-term hydro-meteorological and hydrological assessment period; 2002–2010 represents selected common-period analyses dictated by data availability; and 2017–2019 represents the period of field investigations and stakeholder engagement.

This distinction is consistent with previous research in the basin, where different datasets and analytical periods have been used to investigate specific components of the water balance. For example, Odongo et al. (2019) examined land-use and land-cover effects on evapotranspiration using MODIS and meteorological/reanalysis data for 2003–2012, demonstrating that the appropriate analytical period may vary according to the research variable and data source. The authors found that conversion from grassland to cropland increased evapotranspiration by up to approximately 12%, highlighting the importance of considering both climate and land-use changes when interpreting long-term changes in basin water resources.

Temperature Trends

Temperature analysis undertaken in this study indicates a **persistent warming tendency** within the Lake Naivasha Basin. Mean annual temperature increased at an estimated rate of approximately $0.02^{\circ}\text{C yr}^{-1}$ ($0.2^{\circ}\text{C decade}^{-1}$) during the most recent three decades examined. The observed increase indicates a progressive warming of the basin and is consistent with broader evidence of rising temperatures across Kenya and East Africa. National-scale analyses have reported statistically significant increases in mean, maximum, and minimum temperatures over Kenya, although the magnitude and statistical significance of trends vary spatially and seasonally (Ayugi et al., 2019). More recent climate assessments by the Kenya Meteorological Department likewise indicate a persistent national warming tendency, with warming becoming particularly pronounced after the early 2000s (Kenya Meteorological Department [KMD], 2026).

The results further indicate that **maximum temperatures have increased more rapidly than minimum temperatures**, suggesting that daytime thermal conditions may be changing more strongly than nighttime conditions. Such changes are hydrologically important because increasing air temperatures can influence evaporative demand, crop water requirements, soil-moisture losses, and lake-water losses. However, temperature effects on evaporation should not be interpreted independently of other climatic controls, including humidity, wind speed, net radiation, and precipitation. Indeed, research in the Lake Naivasha Basin has demonstrated that changes in evapotranspiration reflect interactions among climatic conditions and land-use/land-cover transitions rather than temperature alone (Odongo et al., 2019).

The observed warming trend therefore has important implications for the basin's water balance. Higher temperatures may increase atmospheric evaporative demand and potentially intensify water requirements for irrigated agriculture and other water-dependent activities. In a basin already experiencing substantial pressure from agricultural intensification and water abstraction, increasing thermal conditions may contribute to additional stress on available water resources. Odongo et al. (2015) showed that changes in lake volume and river flows in the basin cannot be explained solely by precipitation variability, emphasising the need to consider multiple interacting climatic and anthropogenic drivers. Similarly, Odongo et al. (2019) demonstrated that land-use transitions, particularly expansion of cultivated land, can substantially modify evapotranspiration and consequently influence basin water availability.

The temperature findings should therefore be interpreted as part of a broader **hydro-meteorological system** in which warming interacts with precipitation variability, evaporation, river inflows, land-use change, water abstraction, and lake-level dynamics. The observed warming tendency provides an important indication of changing climatic conditions, but its implications for water availability depend on how temperature changes interact with other components of the basin water balance. Consequently, the temperature results provide an important basis for assessing future water demand, evaporation losses, agricultural water requirements, ecosystem water needs, and the capacity of existing water-governance arrangements to respond to changing hydro-meteorological conditions.

Figure 3(a): Temperature Trends-LNB

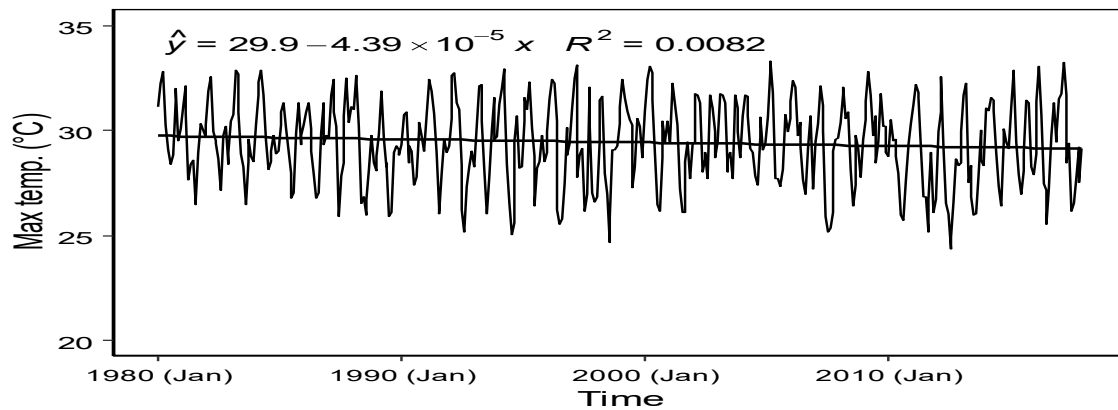


Figure 3(b): Temperature Trends-LNB

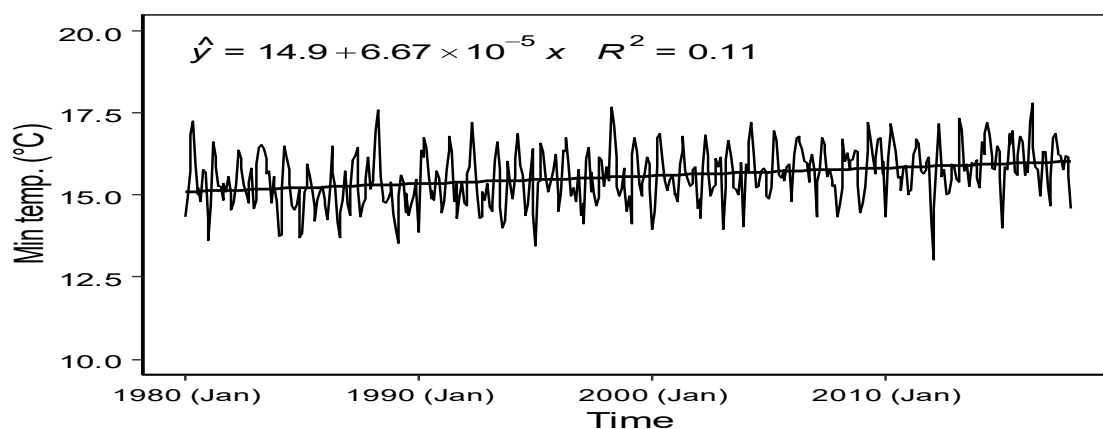
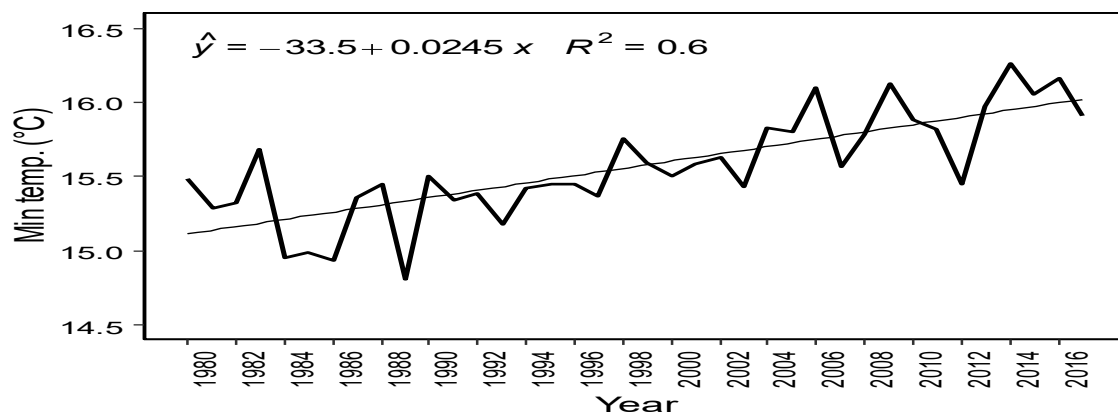


Figure 3(c): Temperature Trends-LNB



Patterns and Trends

Rainfall across the basin exhibits substantial spatial and temporal variability, reflecting the bimodal rainfall regime with long rains from March to May and short rains from October to December. Mann-Kendall trend analyses indicate weak or inconsistent trends in most areas, though some lower-altitude stations show a modest

decline in annual rainfall (~2.3 mm per year), while upper catchments exhibit little or no significant change. High inter-annual variability further complicates trend detection. These patterns suggest that rainfall alone cannot fully explain observed hydrological changes, emphasising the influence of human activities and catchment processes.

SPI/SSI and Extreme Event Analysis

Meteorological and hydrological extremes were assessed using the Standardised Precipitation Index (SPI) and Standardised Streamflow Index (SSI). To capture both short-term and long-term hydroclimatic variability, SPI and SSI were computed at accumulation periods of 1-, 3-, 6-, and 12-month timescales. The 1–3-month windows were used to assess short-term moisture deficits and seasonal anomalies, while the 6–12-month windows were applied to evaluate prolonged drought and hydrological persistence relevant to water resource governance in the Lake Naivasha Basin.

Extreme drought and wet events were classified using standard SPI/SSI threshold categories. Drought conditions were identified when SPI or SSI values fell below -1.0 , while flood/wet conditions were identified when index values exceeded $+1.0$. Event severity was interpreted as follows:

- Mild drought/wet event: ± 1.0 to ± 1.49
- Moderate event: ± 1.5 to ± 1.99
- Severe to extreme event: $\geq \pm 2.0$

A drought or flood event was considered to begin when SPI/SSI crossed the respective threshold (≤ -1.0 or $\geq +1.0$) and ended when values returned above or below the threshold. These thresholds enabled consistent identification, comparison, and interpretation of hydroclimatic extremes across the 2002–2010 analysis period.

Streamflow and Lake Level Variability

River discharge data from major inflows (Malewa and Gilgil Rivers) indicate a general decline in streamflow, particularly in minimum flows, while mean and maximum flows show weak or no consistent trends. The Standardised Streamflow Index (SSI) highlights notable periods of hydrological drought during the 1990s and early 2000s, coinciding with low lake levels. Lake Naivasha exhibits substantial interannual fluctuations, largely driven by variable inflows and evaporation. Despite occasional rises due to extreme rainfall events, the long-term trend

indicates declining lake levels, posing risks to aquatic ecosystems and water availability.

Evapotranspiration and Water Balance

Rising temperatures and extended dry periods have led to an estimated 5–8% increase in evapotranspiration (ET) over the study period. This has exacerbated water losses from the lake and catchment, amplifying the effects of reduced rainfall and streamflow. Water balance analyses suggest that declining inflows combined with increasing ET are primary drivers of observed lake level reductions. While groundwater inflows and outflows remain less understood, they likely play a critical role in buffering the system during dry spells.

Impact of Land Use Changes

Remote sensing and land cover analyses reveal extensive anthropogenic alterations in the basin over the past two decades. Significant conversion of natural forests and wetlands to agriculture and urban areas has occurred, particularly in upper catchments. Deforestation likely increases surface runoff and sediment loads, reducing water quality and altering river hydrology. Agricultural expansion, notably floriculture near the lake, has intensified water abstraction, further stressing the lake's water balance. Land cover changes directly influence runoff, infiltration, and evapotranspiration, underscoring the need for integrated catchment management. Protecting wetlands, riparian zones, and upland areas is critical to maintaining hydrological resilience. Ecosystem services, including fish and bird habitats and tourism, depend on stable water levels and quality. Governance strategies must emphasise sustainable water use, pollution control, and flexible management approaches to buffer against both drought and flood risks. Scenario planning for drier or wetter futures is essential, as even small shifts in effective rainfall (0.4–2%) can significantly alter lake levels.

Integrating Climate and Human Impacts

While rainfall trends are weak or spatially heterogeneous, lake volumes and streamflow patterns exhibit stronger changes, suggesting that

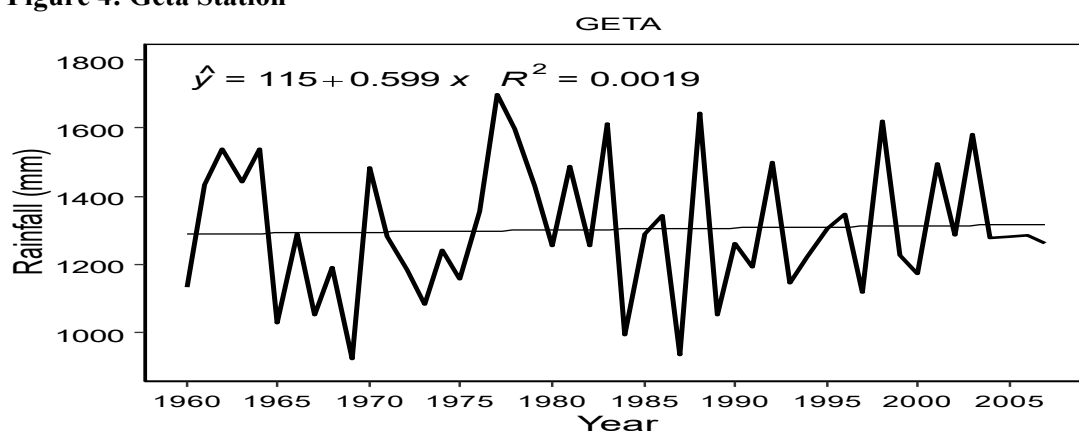
non-climatic factors—land use change, water abstraction, wetland loss, and groundwater dynamics—are dominant drivers of hydrological variability. Recent modest increases in lake area/levels demonstrate that the system responds sensitively to small changes in inflow or evaporation, emphasising the importance of adaptive management.

The work of Odongo et al. (2015) supports this interpretation, indicating that no clear evidence of climate change impacts has been detected in the

basin over the past 50 years, highlighting the primary role of human activities in shaping hydrological trends.

While these declines raise concerns about reduced water availability during dry periods potentially exacerbating drought conditions and impacting downstream water resources overall, long-term rainfall trends remain weak or ambiguous, underscoring the high variability and complexity of the basin’s precipitation patterns.

Figure 4: Geta Station



The mismatch between rainfall trends (weak or none) vs lake-volume decline suggests that changes are not solely explained by rainfall changes. Odongo et al. conclude that “no evidence of any impact of climate change for the Lake Naivasha basin over the past 50 years” – meaning not yet detectably linked to climate forcing in their analysis. Other factors such as land use change (deforestation, agricultural expansion), increased

abstraction (horticulture, domestic water supply), changes in runoff/evapotranspiration and groundwater behaviour seem to play a strong role. The more recent upward trend in lake area/levels suggests that modest increases in rainfall (or reductions in evaporation) or changes in catchment behaviour can produce significant lake response — a small change in effective rainfall (~0.4–2 %) can lead to the observed rise.

Figure 5(a): Kongoni Station

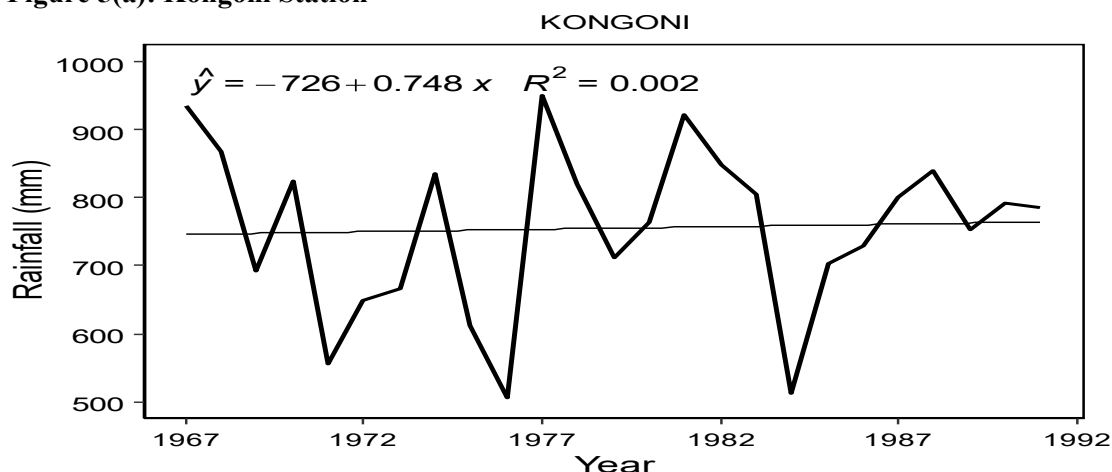
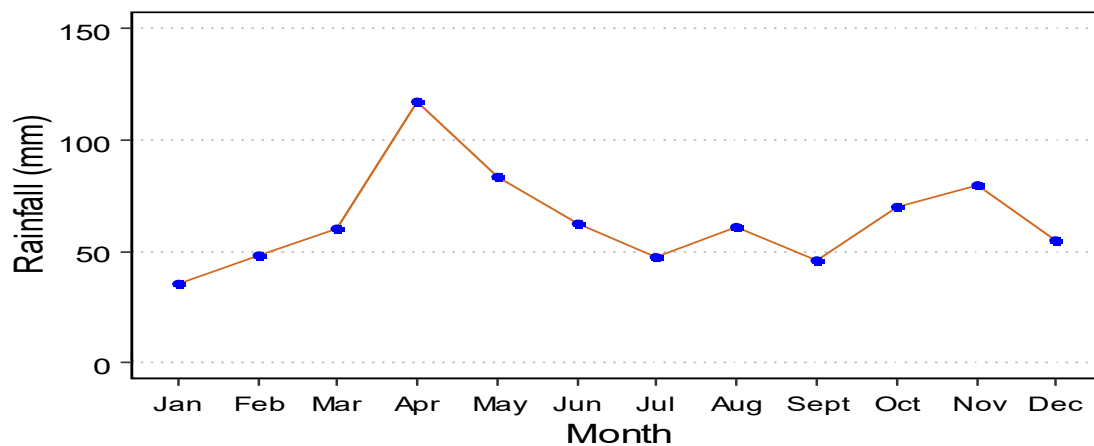


Figure 5(b): Kongoni Station



Spatial variability is a key feature of the basin's rainfall regime. Upland areas, such as Kinangop, receive substantially higher rainfall than the lowland lake zone, making highland contributions particularly important for sustaining streamflow and lake levels. The bimodal rainfall pattern—long rains from March to June and short rains from October to December remains the dominant seasonal regime, shaping both surface runoff and water availability.

Despite weak trends in rainfall, other hydro-meteorological variables, including minimum river flows and lake volumes, display more pronounced changes. This suggests that non-climatic drivers, including land use change, agricultural expansion, water abstraction, wetland

loss, and alterations in groundwater dynamics, are likely exerting a stronger influence on the basin's hydrology. The absence of a strong, statistically significant monotonic trend in annual rainfall over much of the 1960–2010 period reinforces the conclusion that rainfall alone cannot explain long-term changes in lake levels and streamflow, a point highlighted by Odongo et al. (2015). These findings underscore the importance of integrating climate and human-induced factors when assessing hydrological trends in the Lake Naivasha Basin. Understanding the combined effects of variable rainfall, catchment management, and water use is essential for developing sustainable water resource strategies and maintaining ecosystem resilience.

Figure 6(a): Geta Station

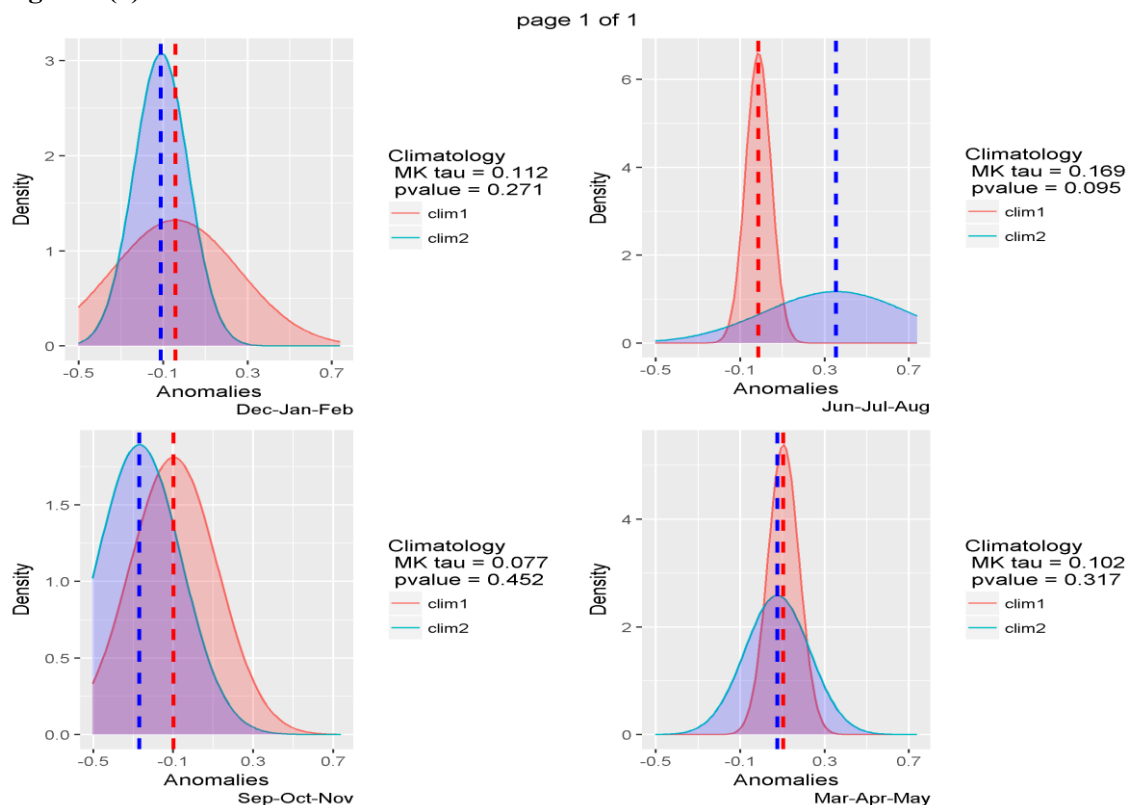


Figure 6(b): Naivasha Water Baillif Station

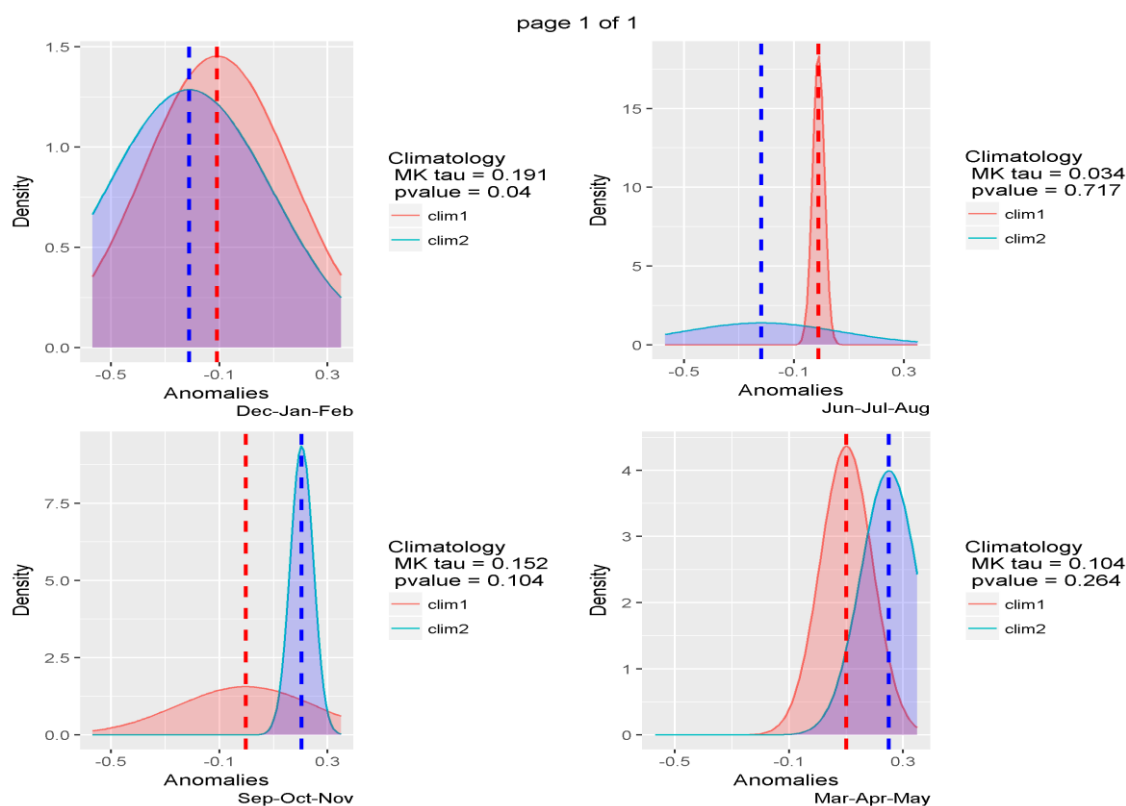


Figure 6(c): New KCC Naivasha Station

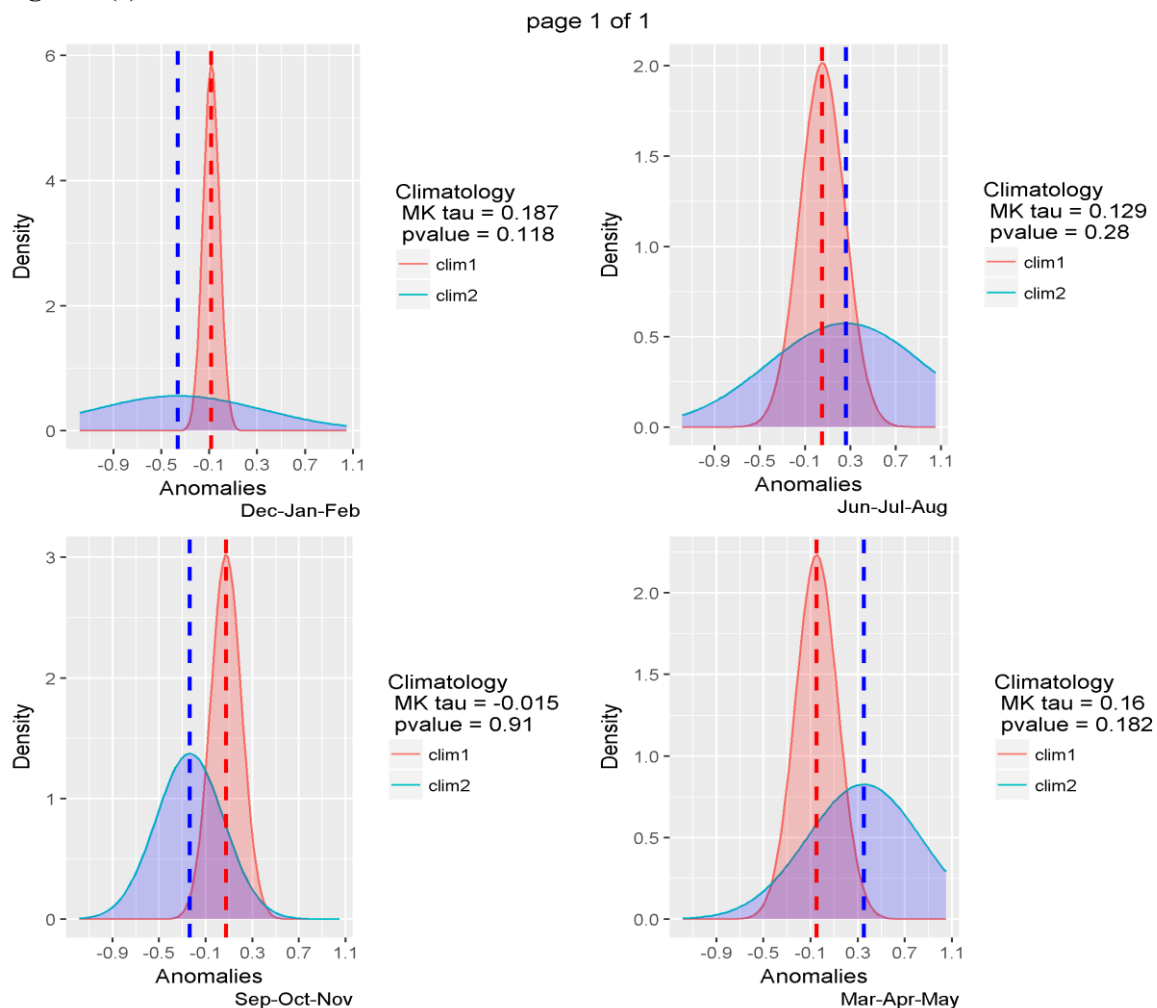


Figure 7(a): Ts- Animal Husbandry Station, (Naivasha (KALRO, Naivasha))

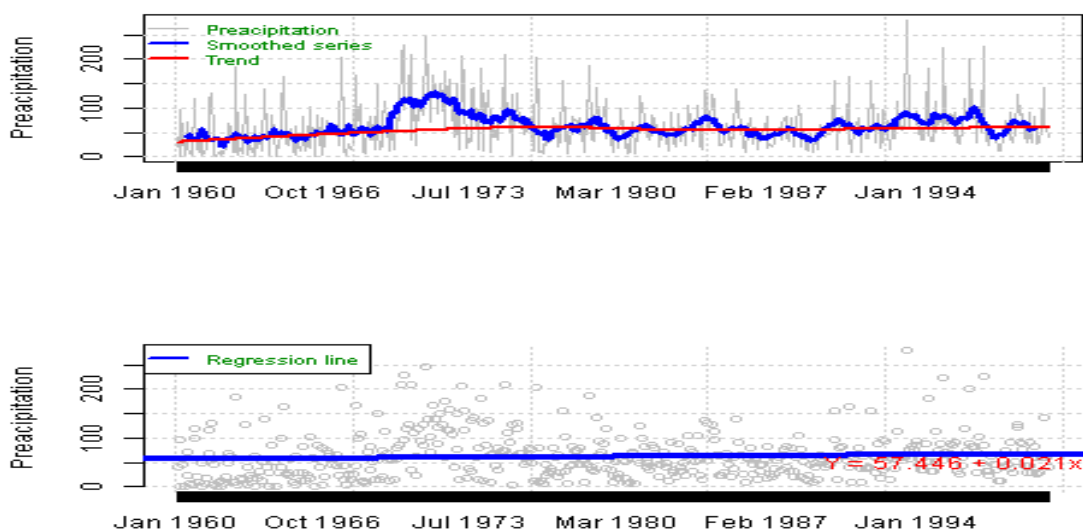
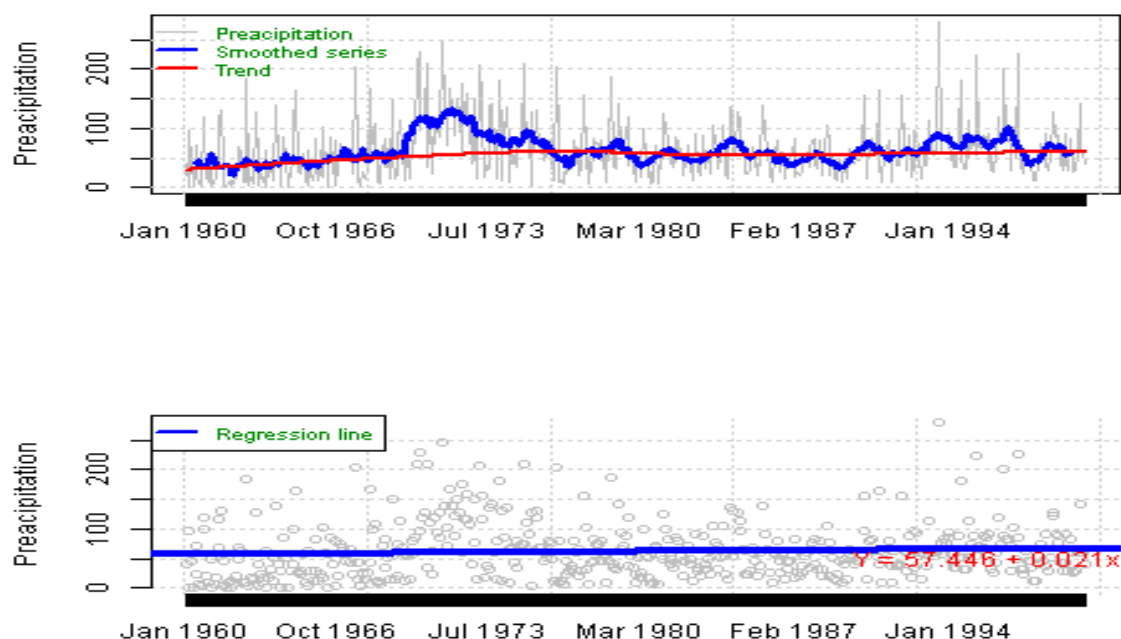


Figure 7(b): Ts New KCC Naivasha Station



Lake Naivasha Water Levels Analysis

The Lake Naivasha Basin exhibits a relatively tight water balance, characterised by high evaporation relative to rainfall. Consequently, the lake, tributaries, and groundwater resources are highly sensitive to fluctuations in rainfall timing and magnitude, as well as changes in evaporation. Declines in lake volume or drops in minimum river flows reduce the system's buffering capacity during dry seasons, compromising the reliability of water supply for irrigation, domestic use, and ecosystem services.

Between approximately 2002 and 2010, Lake Naivasha experienced substantial shrinkage, with water levels dropping at an estimated rate of ~10 cm per year. This trend heightened the risk of water stress during dry periods. More recently, rising lake levels suggest temporarily improved water availability; however, the underlying drivers whether increased rainfall, catchment changes, or shifts in groundwater dynamics remain uncertain, making reliance on this apparent improvement potentially risky.

Seasonal rainfall variability, particularly during the long rains (March–June) and short rains (October–December), directly affects dry-season flows. Observations indicate that some tributaries, such as the Karati River, occasionally dry up during dry periods. Conversely, elevated lake levels increase the risk of flooding and shoreline inundation, particularly given the lake's shallow depth and large surface area fluctuations. Most documented publications highlight frequent flooding events that have displaced local populations. These hydrological extremes, such as low flows and droughts to high water levels and floods increases the significant challenges for water management infrastructure, ecosystem resilience, and livelihoods dependent on the lake, including fisheries, tourism, and irrigation. Effective management must therefore account for both climatic variability and human-induced changes to ensure sustainable and resilient water resource use in the basin.

Figure 8(a): Long-term Lake Naivasha Level Trends

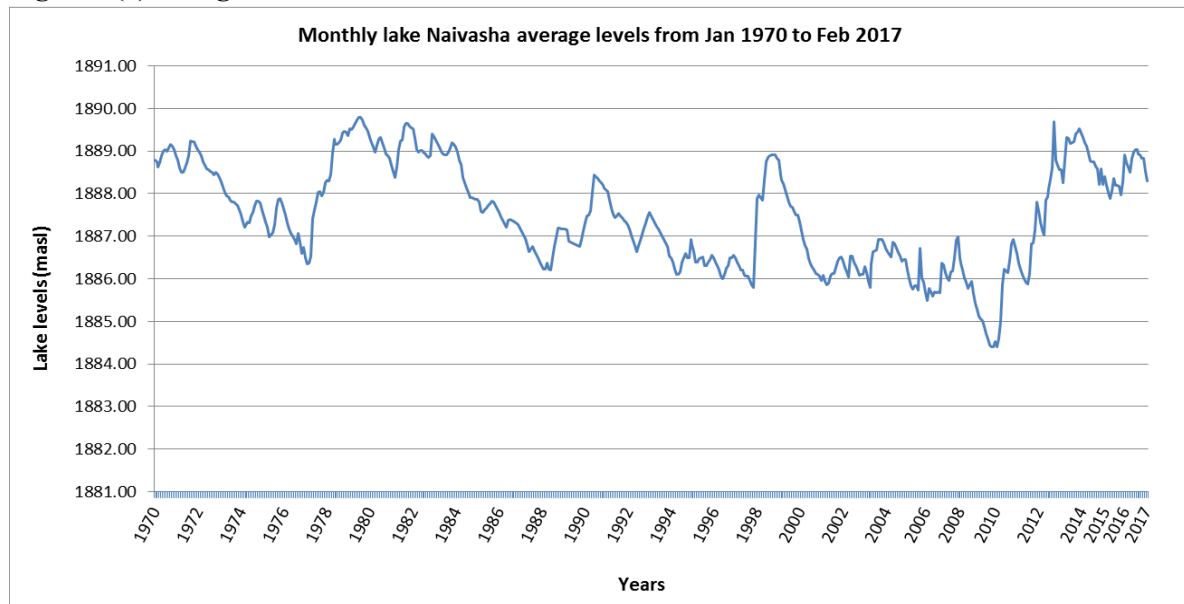


Figure 8(b): Long-term Lake Naivasha Level Trends

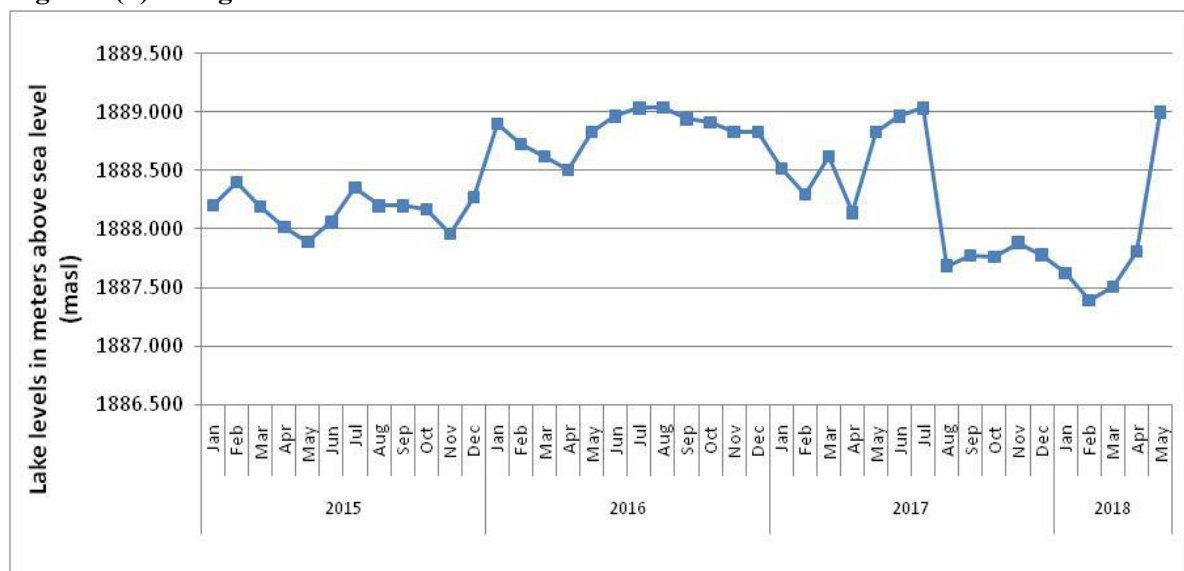
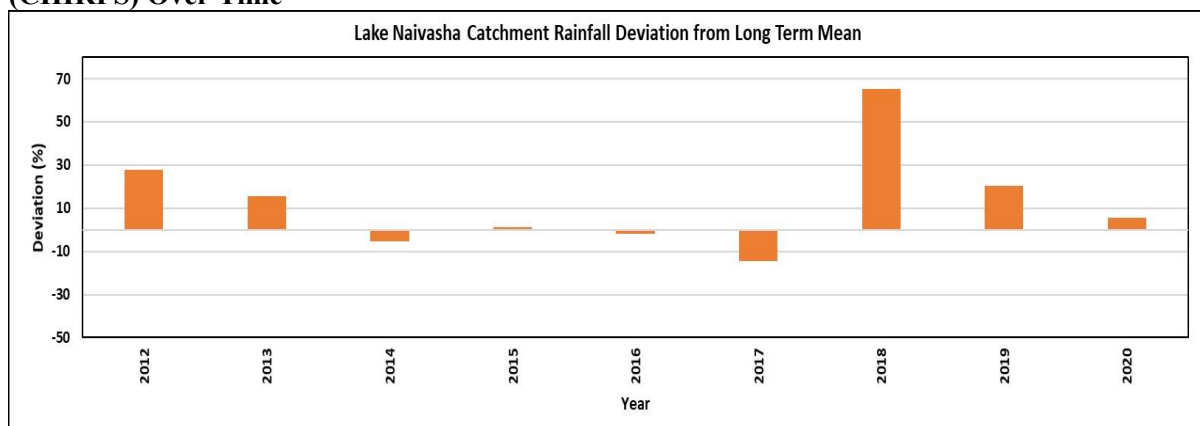


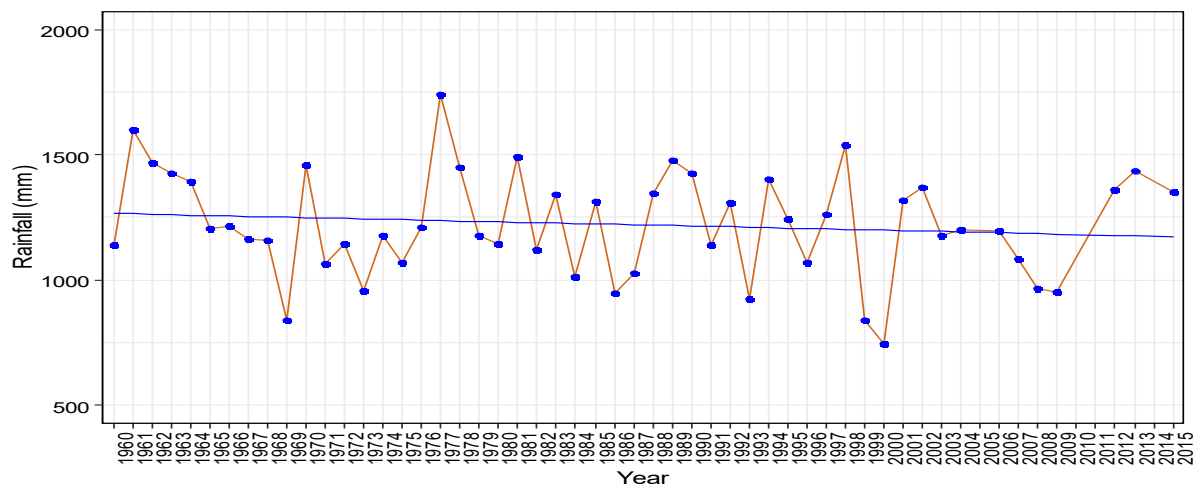
Figure 9: Evolution of Lake Naivasha's Surface Area, Compared to Basin (catchment) rainfall (CHIRPS) Over Time



Source: (Herrnegger, M. et al., 2021; RCMRD, 2021)

RAINFALL TRENDS FOR SELECTED STATIONS IN LNB OVER THE YEARS

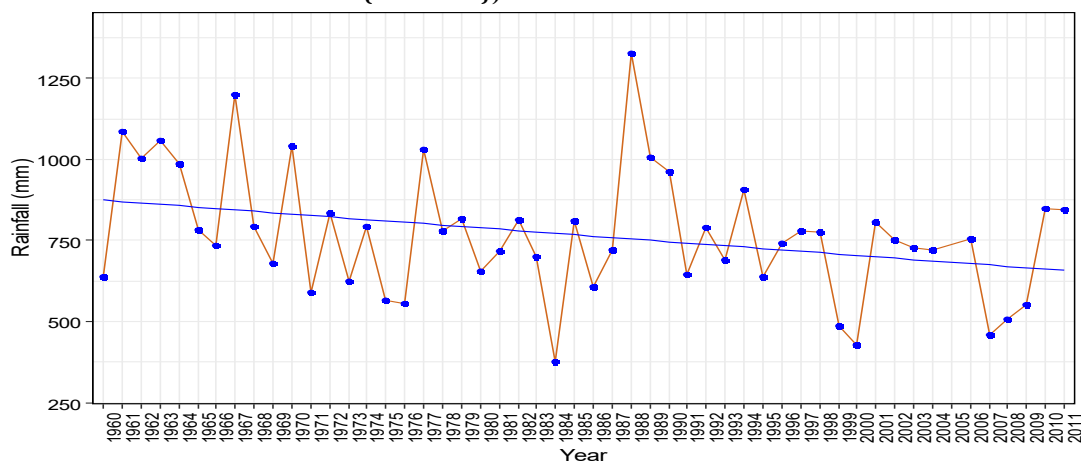
Figure 10: Kinangop Station



The rainfall graph for Kinangop depicts Rainfall Variability trends from 1960–2018, with a fluctuating pattern significantly from year to year. The highest rainfall values reach around 1800–1900 mm, while the lowest drop below 800 mm, suggesting high inter-annual variability, meaning rainfall is irregular and unpredictable. This shows that the trendline is slightly downward, indicating a small long-term decrease in rainfall over the study period (1960–2018; the decrease is not steep but shows a gradual reduction in average annual rainfall. The graph depicts a periodic pattern such that the 1960s–1970s: Mostly high rainfall years, with values above 1200 mm. 1980s–1990s: Noticeable fluctuations—some very wet years and

some very dry ones. 2000s–2010s: Rainfall becomes more stable but overall lower compared to earlier decades. The interpretation follows that over the last six decades, the area experienced a slight decline in rainfall and increased variability. The irregular ups and downs indicate that rainfall is inconsistent, which could be related to climate, such as El Niño–Southern Oscillation (ENSO) events, changes in regional climate patterns, or human impacts on the environment. Despite some years of heavy rainfall, the long-term tendency shows a gradual drying trend due to climate variability, which is a concern for water resource management in LNB.

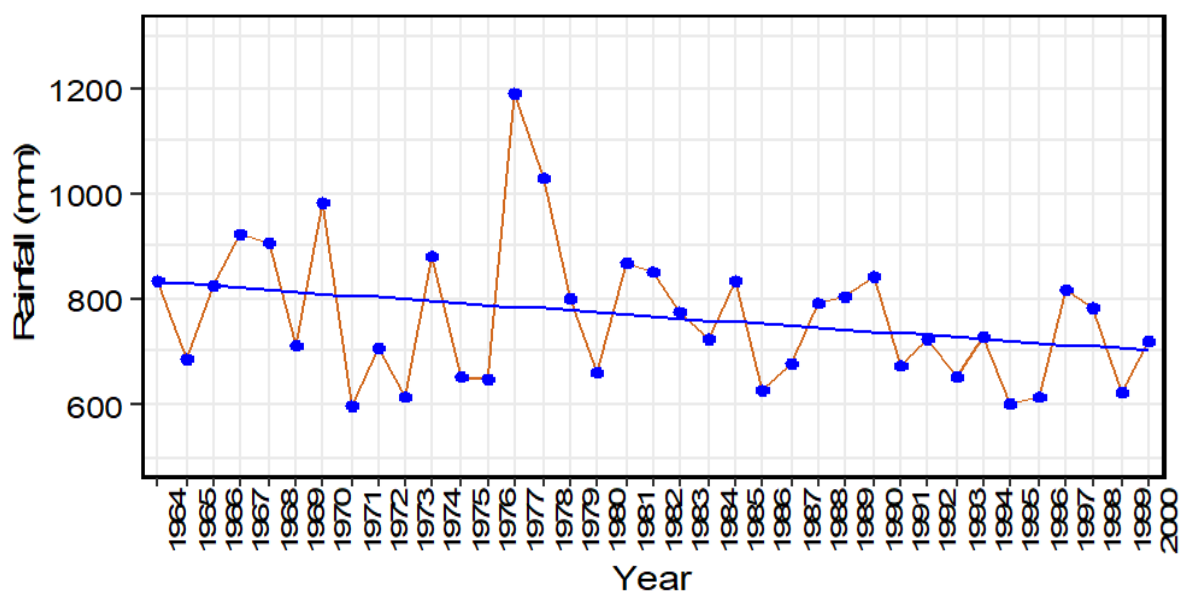
Figure 11: National Animal Husbandry (NAH) Research Station (now Kenya Agricultural and Livestock Research Institute {KALRO})



Observations from the NAH Research rainfall graph shows that there has been rainfall variability, rainfall values fluctuate significantly from year to year, with high peaks (around 1962, 1973, 1985, and 1995) where rainfall exceeds 1000 mm. and low points (around 1974, 1983, 2005) where rainfall drops below 500 mm. The trendline slopes slightly downward, suggesting a gradual decline in annual rainfall over the years. This means that, on average, the area has been receiving less rainfall over time, suggesting climate variability and change. The graph indicates a long-term decreasing trend in rainfall from the 1960s to the 2010s. The fluctuations

show that rainfall is highly irregular, implying climatic variability (possibly linked to El Niño/La Niña or local climate shifts). Despite some wet years, the general tendency is towards reduced precipitation. The graph implies that decreasing rainfall could reduce crop yields and rainfed farming. Declining rainfall may lead to lower groundwater recharge and reduced surface water availability, increasing the risk of drought, vegetation stress, and soil erosion. Therefore, there is a need for Water conservation measures, improved irrigation, and sustainable agricultural practices.

Figure 12: Olaragwi Farm Station



The graph shows the Olaragwi rainfall station; the x-axis represents the years (ranging roughly from 1960 to 1991). The y-axis represents the rainfall in millimetres (mm). The blue line with dots represents the annual rainfall values for each year. There's also a trend line (in darker blue), showing the general direction, meaning this area experienced fluctuating rainfall patterns with no consistent stability.

Over the three decades shown, there appears to be a long-term decline in rainfall. Such a pattern could indicate changing climatic conditions or increasing aridity in the region. Possible implications for agricultural productivity could be unreliable rainfall that affects crop productivity, a

potential decrease in available water for domestic and industrial use that also leads to droughts, soil degradation, and ecosystem stress in the Lake Naivasha basin.

Lake Naivasha Basin Stakeholder Analysis

Stakeholders in the Lake Naivasha Basin play complementary and interdependent roles in water management and governance. These include;

- Government agencies provide regulatory and policy frameworks.
- Communities and WRUAs ensure localised compliance, equity, and environmental stewardship.

- Private sector drives innovation, efficiency, and co-financing of conservation.
- NGOs and research institutions strengthen capacity, monitoring, and adaptive management.
- Multi-stakeholder coordination platforms (like Imarisha Naivasha Trust) integrate these efforts toward sustainable basin governance.

The success of water management in the Lake Naivasha Basin depends on inclusive, transparent,

data-driven, and adaptive governance, recognising that climate variability, land-use pressure, and socio-economic growth are interconnected challenges requiring collective action. The five categories of key stakeholders in this study included Government Institutions (National and County); National and International NGOs/Organizations; Local Civil Society Organisations; Private Sector Organisations; and Universities and Research Organisations working together to enhance sustainability and water Governance.

Table 3: Showing Stakeholder Engagement in LNB

Governance Level	Stakeholder Example	Core Function
National	MoWSI, WRA, NEMA	Policy formulation, regulation, national oversight
Regional/County	Nakuru & Nyandarua County Govts	Implementation, land & water-use management
Catchment/Sub-Catchment	WRUAs, Imarisha Naivasha Trust	Day-to-day water management, coordination
Community	CBOs, local users, smallholders	Resource stewardship, local compliance
Private Sector	Growers, geothermal, tourism	Efficient use, co-financing conservation
Research/NGOs	Universities, WWF	Data, innovation, awareness & advocacy

Implications and Future Outlook

The Lake Naivasha Basin is characterised by a bimodal rainfall regime, moderate annual precipitation, high evaporative demand, and pronounced spatial climatic variability driven by topography. Long-term hydro-meteorological observations indicate substantial temporal variability in rainfall, river discharge, and lake water levels. Although Lake Naivasha experienced prolonged declines in water levels over several decades, a period of recovery has been observed since approximately 2010. These fluctuations reflect the combined influence of climatic variability and anthropogenic drivers, including land-use change, increasing water abstraction, catchment degradation, and changing hydrological connectivity.

The study demonstrates that climate variability, particularly changes in rainfall distribution, increasing temperatures, and enhanced

evapotranspiration, is intensifying pressure on the basin's water resources. These climatic stresses are compounded by rapid agricultural expansion, urbanisation, and growing water demand, resulting in reduced streamflows, altered groundwater recharge, and increasing competition among domestic, agricultural, industrial, and environmental water users. Consequently, the basin's hydrological regime is increasingly shaped by the interaction between climate and human activities rather than by climatic factors alone. These findings underscore the need for an integrated and adaptive water governance framework that recognises the interdependence of climate, hydrology, ecosystems, and socio-economic development. Sustainable management of the Lake Naivasha Basin will require coordinated policies that integrate climate adaptation with land-use planning, water allocation, ecosystem conservation, and catchment restoration.

Priority policy actions should include strengthening hydro-meteorological monitoring networks and data management systems; integrating land-use planning with water resources governance; regulating surface and groundwater abstractions based on adaptive allocation rules that respond to climatic variability; protecting groundwater recharge areas, riparian corridors, and wetland buffer zones; restoring degraded catchments to enhance hydrological resilience; and improving water-use efficiency across all sectors. In addition, scenario-based planning for both drought and flood risks should be incorporated into basin management to enhance preparedness under future climate uncertainty.

Overall, the study highlights that sustainable water security in the Lake Naivasha Basin will depend on transitioning from reactive resource management to an evidence-based, climate-resilient, and ecosystem-centred governance approach capable of addressing the combined impacts of climate variability and increasing anthropogenic pressures. Such an integrated framework is essential for safeguarding the basin's ecological integrity while sustaining the livelihoods and economic activities that depend on its water resources.

REFERENCES

- Alexandersson, H. (1986). A homogeneity test applied to precipitation data. *Journal of Climatology*, 6(6), 661–675. <https://doi.org/10.1002/joc.3370060607>
- Becht, R., & Harper, D. M. (2002). Towards an understanding of human impact upon the hydrology of Lake Naivasha, Kenya. *Hydrobiologia*, 488, 1–11. <https://doi.org/10.1023/A:1023318007715>
- Kyambia, M. M., & Mutua, B. M. (2014). Detection of trends in extreme streamflow due to climate variability in the Lake Naivasha basin, Kenya. *Journal of Water and Climate Change*. <https://doi.org/10.1080/15715124.2014.967255>
- Midekisa, A., Holl, F., Savory, D. J., Andrade-Pacheco, R., Gething, P. W., Bennett, A., & Sturrock, H. J. (2017). Mapping land cover change over continental Africa using Landsat and Google Earth Engine cloud computing. *PLOS ONE*, 12(9), e0184926. <https://doi.org/10.1371/journal.pone.0184926>
- Ndungu, J. N., Ogada, J. O., & van der Veen, A. (2014). Supporting IWRM through spatial integrated assessment in the Lake Naivasha basin, Kenya. *International Journal of Water Resources Development*, 30(3), 605–618. <https://doi.org/10.1080/07900627.2014.920248>
- Odongo, V. O., van der Tol, C., van Oel, P. R., Meins, F. M., Becht, R., Onyando, J. O., & Su, Z. (2015). Characterisation of hydroclimatological trends and variability in the Lake Naivasha basin, Kenya. *Hydrological Processes*, 29(15), 3276–3293. <https://doi.org/10.1002/hyp.10443>
- Odongo, V. O., van Oel, P. R., van der Tol, C., & Su, Z. (2019). Impact of land use and land cover transitions and climate on evapotranspiration in the Lake Naivasha Basin, Kenya. *Science of the Total Environment*, 682, 19–30. <https://doi.org/10.1016/j.scitotenv.2019.04.062>
- Pettitt, A. N. (1979). A non-parametric approach to the change-point problem. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 28(2), 126–135. <https://doi.org/10.2307/2346729>
- Yihdego, Y., Reta, G., & Becht, R. (2016). Hydrological analysis as a technical tool to support strategic and economic development: A case study of Lake Naivasha, Kenya. *Water and Environment Journal*, 30(1–2), 40–48. <https://doi.org/10.1111/wej.12162>